Clustering Concepts

Feature Extraction and Feature Selection

Techniques used in machine learning to improve model performance by handling irrelevant or redundant features

Feature Extraction

- Transforms existing features to better capture underlying patterns
- Useful when data is high dimensional or complex

Feature Selection

- Selects most relevant features and throws away the rest
- Reduces overfitting and increases accuracy

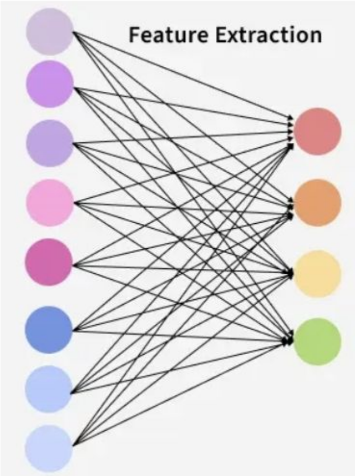


Figure 1. Feature Extraction (Geeksfor Geeks, 2025)

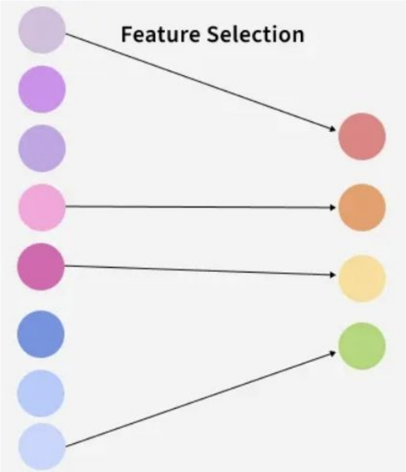


Figure 2. Feature Selection (Geeksfor Geeks, 2025)

What is Clustering?

- Clustering finds structure by grouping similar objects together
- A good cluster has strong intra-cluster cohesion, well-separation, and is interpretable
- The Silhouette Score measures how well a point fit compared to others
- The Dunn Index evaluates cluster separation

Different Types of Clustering

Partitioning methods

- K-Means

Hierarchical methods

- Divisive Clustering (DIANA)
- Agglomerative clustering (AGNES)

Density based method

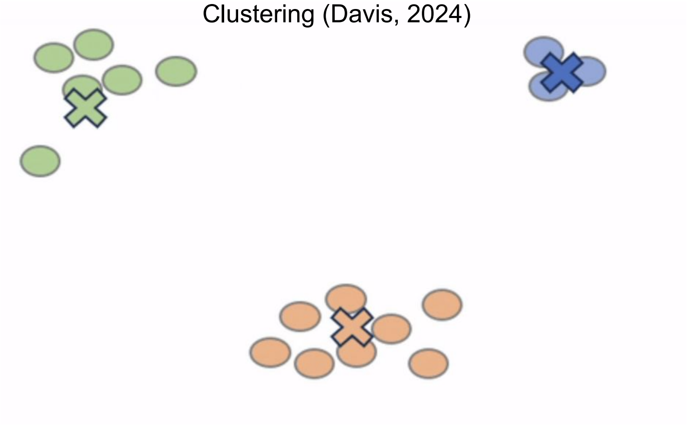
- DBSCAN

Distribution based

- Gaussian Mixture Model (GMM)

K-Means Clustering

Figure 3. Centroid-Based Clustering (Davis, 2024)



How it works: Iteratively assigns points to closest centroid and recalculates the mean

Best for:

- Continuous numerical data on the same scale
- No massive outliers
- Regularly-shaped clusters
- Number of clusters is known
  - Elbow Method or Silhouette Score

Figure 4. Elbow Method vs. Silhouette Score comparison graph (GeeksforGeeks, 2025)

Hierarchical Methods

How they work: Creates tree like structures by iteratively merging (AGNES) or splitting (DIANA) groups

Best for:

- Capture a hierarchy or visualize relationships between groups
  - Dendrogram
- Number of clusters is unknown
- The dataset is a nested, tree like structure
- The dataset is small
  - Computationally intensive

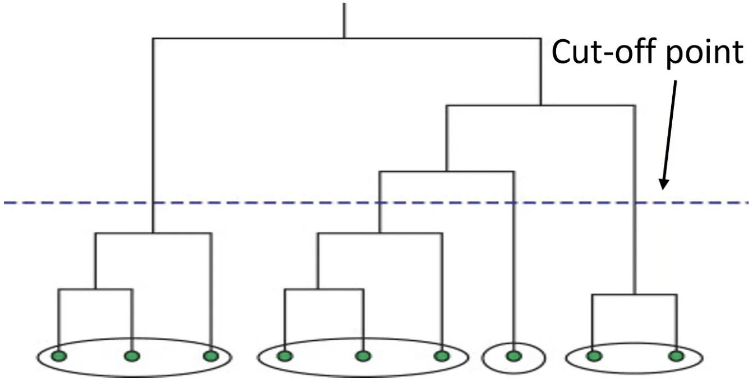


Figure 5. Hierarchical Clustering (Davis, 2024)

DBSCAN Clustering

How it works: Groups data based on spatial density thresholds (minPts and eps)

Best for:

- Number of clusters is unknown
- Irregularly shaped clusters
- Significant noise or outliers

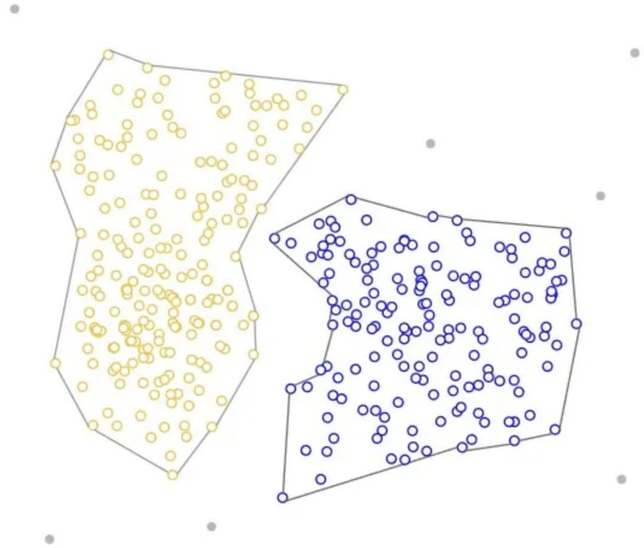


Figure 6. Density-Based Clustering (Davis, 2024)

Gaussian Mixture Model (GMM)

Most datasets can be described by a Gaussian Distribution

How it works: Calculates the probability of a point belonging to overlapping normal distributions

Best for:

- Overlapping or unclear boundaries
- Varying cluster shapes and sizes
- Data is normally distributed

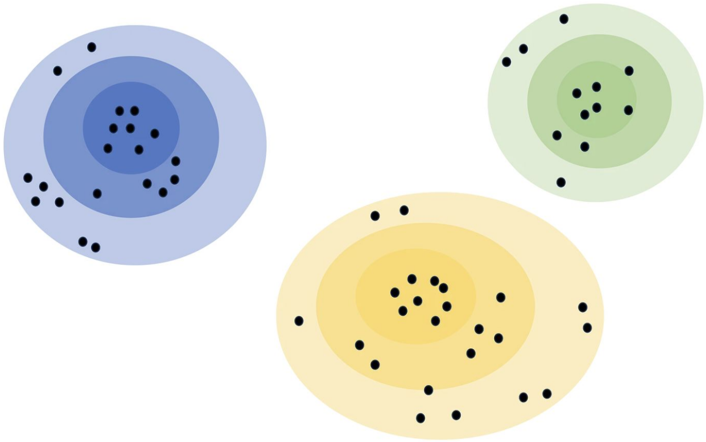
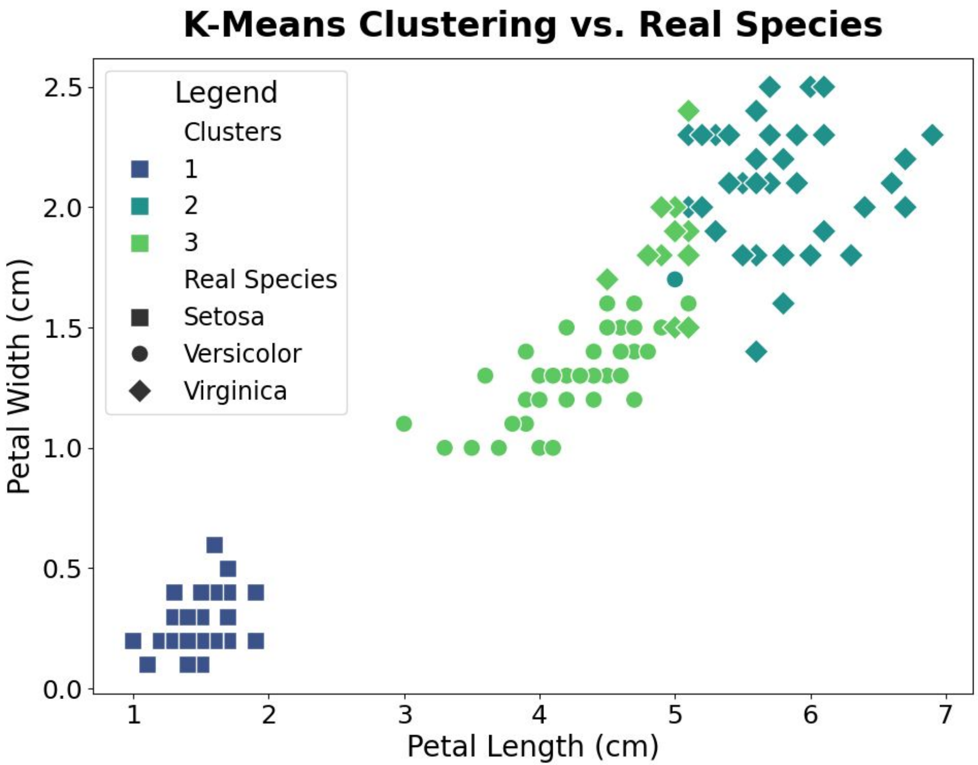


Figure 7. Distribution-Based Clustering (Davis, 2024)

Clustering the Iris Dataset Using K-Means



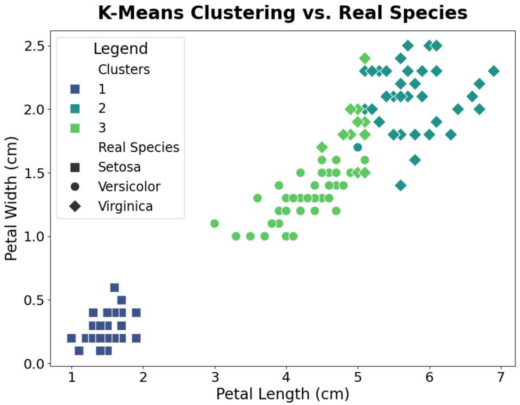
Contains:

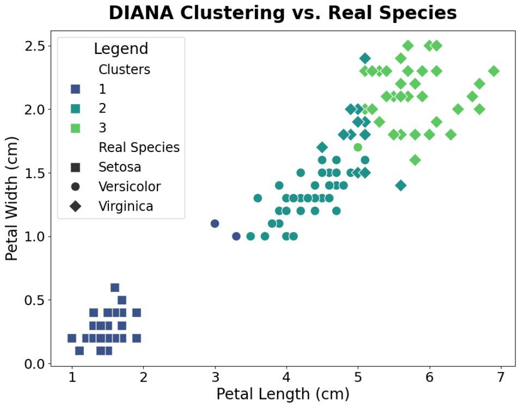
- Class
- Petal Width and Length
- Sepal Width and Length

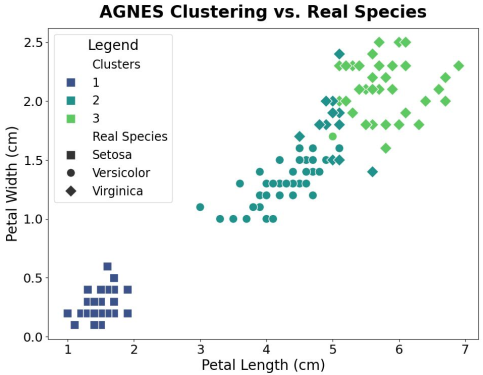
- Did well clustering Setosa
- Had trouble with Versicolor and Virginica

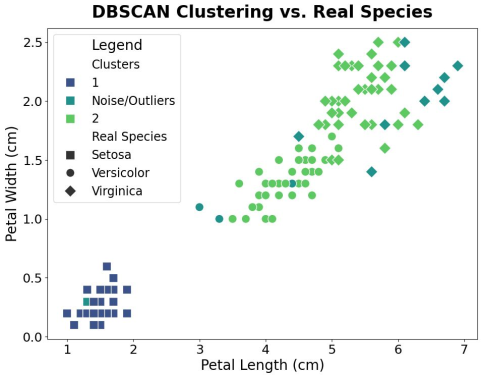
Iris Dataset (Fisher, 1936)

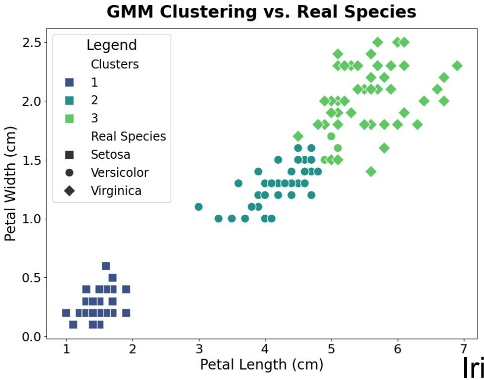
Clustering the Iris Dataset Using Different Algorithms











Iris Dataset (Fisher, 1936)

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